# FULL TEST – IV

**Time Allotted: 3 Hours** **Maximum Marks: 432**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

## INSTRUCTIONS

### A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

### B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

### C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 03 and 10 to 12)** contains 6 multiple choice questions which have only one correct answer. Each question carries **+8 marks** for correct answer and **– 2 mark** for wrong answer.

**Section-A (04 to 09 and 13 to 30)** contains 24 multiple choice questions which have only one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

| | |
|---|---|
| **Name of the Candidate** | |
| **Enrolment No.** | |

ALL INDIA TEST SERIES

FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

## Useful Data

### PHYSICS

| | |
|---|---|
| Acceleration due to gravity | $g = 10\ m/s^2$ |
| Planck constant | $h = 6.6 \times 10^{-34}$ J-s |
| Charge of electron | $e = 1.6 \times 10^{-19}$ C |
| Mass of electron | $m_e = 9.1 \times 10^{-31}$ kg |
| Permittivity of free space | $\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$ |
| Density of water | $\rho_{water} = 10^3\ kg/m^3$ |
| Atmospheric pressure | $P_a = 10^5\ N/m^2$ |
| Gas constant | $R = 8.314\ J\ K^{-1}\ mol^{-1}$ |

### CHEMISTRY

| | | | |
|---|---|---|---|
| Gas Constant | R | = | $8.314\ J\ K^{-1}\ mol^{-1}$ |
| | | = | $0.0821$ Lit atm $K^{-1}\ mol^{-1}$ |
| | | = | $1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$ |
| Avogadro's Number | $N_a$ | = | $6.023 \times 10^{23}$ |
| Planck's constant | h | = | $6.625 \times 10^{-34}$ J·s |
| | | = | $6.625 \times 10^{-27}$ erg·s |
| 1 Faraday | | = | 96500 coulomb |
| 1 calorie | | = | 4.2 joule |
| 1 amu | | = | $1.66 \times 10^{-27}$ kg |
| 1 eV | | = | $1.6 \times 10^{-19}$ J |

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

# Physics

## PART – I

## SECTION – A
### Single Correct Choice Type

This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. The unit vector perpendicular to $\hat{i}-2\hat{j}+\hat{k}$ and $3\hat{i}+\hat{j}-2\hat{k}$ is

(A) $\frac{5\hat{i}+3\hat{j}+7\hat{k}}{\sqrt{83}}$ (B) $\frac{3\hat{i}+5\hat{j}+7\hat{k}}{\sqrt{83}}$

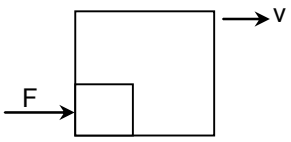
(C) $\frac{5\hat{i}+3\hat{j}-7\hat{k}}{\sqrt{83}}$ (D) $\frac{3\hat{i}-5\hat{j}+7\hat{k}}{\sqrt{83}}$

2. A ball is projected at an angle $60^{o}$ with the horizontal with speed 30 m/s. After certain time it makes an angle $45^{o}$ with the horizontal. The speed of ball at that moment is

(A) 30 m/s (B) $15\sqrt{2}$ m/s

(C) $\frac{15}{\sqrt{2}}$ m/s (D) $30\sqrt{2}$ m/s

3. A block of mass 1 kg rests inside a cubical vessel, as shown in the figure which is moving with a velocity of $v=\sqrt{29}\,t$ m/s, where t is in sec. All the surfaces are smooth. The block is at rest w.r.t. the cube, the total force exerted by the cube on the block




(A) $\sqrt{129}$ N (B) $\sqrt{130}$ N

(C) 10 N (D) $\sqrt{29}$ N

4. A ball falls vertically on to a floor, with momentum p, and then bounces repeatedly, the coefficient of restitution is e. The total momentum imparted by the ball to the floor is

(A) $p(1+e)$ (B) $\frac{p}{(1-e)}$

(C) $p\frac{(1+e)}{(1-e)}$ (D) $p\left(1-\frac{1}{e}\right)$

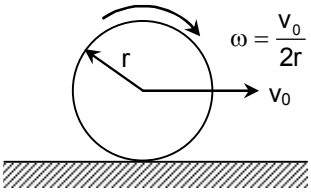
***Space for Rough work***

5. A particle of mass 4m which is at rest explodes into three fragments. Two of the fragments each of mass m are found to move with a speed v each in mutually perpendicular directions along x and y axis. Calculate momentum of the fragment 2m after explosion, the energy released in the process of explosion.

(A) $2m\left(\frac{v}{2}\hat{i}-\frac{v}{2}\hat{j}\right),\ \frac{3}{2}mv^2$ (B) $-2m\left(\frac{v}{2}\hat{i}+\frac{v}{2}\hat{j}\right),\ \frac{3}{2}mv^2$

(C) $-2m\left(\frac{v}{2}\hat{i}+\frac{v}{2}\hat{j}\right),\ \frac{3}{4}mv^2$ (D) $-2m\left(\frac{v}{2}\hat{i}-\frac{v}{2}\hat{j}\right),\ \frac{3}{4}mv^2$

6. A solid sphere of mass M and radius r slips on a rough horizontal surface. At some instant, it has translational velocity $v_0$ and angular velocity $\omega\left(=\frac{v_0}{2r}\right)$ about its center as shown in figure. The translational velocity after the sphere starts pure rolling motion is




(A) $\frac{v_0}{7}$ (B) $\frac{2v_0}{7}$

(C) $\frac{5v_0}{7}$ (D) $\frac{6v_0}{7}$

7. A uniform rod of length l and mass M is suspended from two inextensible string as shown in figure. Tension in left string at the instant, when right string snaps, is



(A) $\frac{mg}{2}$ (B) mg

(C) $\frac{mg}{4}$ (D) $\frac{mg}{8}$

***Space for Rough work***

8. In a certain region at space gravitational field is given by $I = -\frac{k}{r}$. Taking the reference point to be at $r = r_0$ with $V = V_0$, the potential $V_{(r)}$ is given by

(A) $V_{(r)} = K \ln\frac{r}{r_0} + V_0$ (B) $V_{(r)} = K \ln\frac{r}{r_0} - V_0$

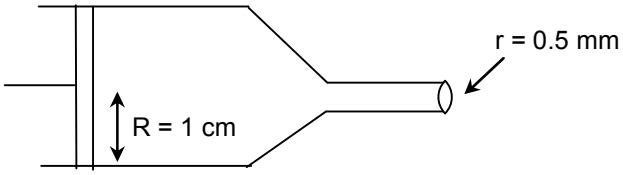
(C) $V_{(r)} = K \ln\frac{r_0}{r} + V_0$ (D) $V_{(r)} = K \ln\frac{r_0}{r} - V_0$

9. A planet of mass m moves along an ellipse around the sun so that its maximum and minimum distances from the sun are equal to R and r respectively. The angular momentum of this planet relative to the centre of the sun (mass M) is

(A) $2\sqrt{\frac{GMmrR}{r+R}}$ (B) $2R\sqrt{\frac{GMmR}{r+R}}$

(C) $2m\sqrt{\frac{GMrR}{r+R}}$ (D) $m\sqrt{\frac{2GMrR}{r+R}}$

10. A piston of a syringe pushes a liquid with a speed of 1 cm/sec. The radii of the syringe tube and the needle are R = 1 cm and r = 0.5 mm respectively. The velocity of the liquid coming out of the needle is




(A) 2 cm/sec. (B) 20 cm/sec.
(C) 10 cm/sec. (D) 400 cm/sec.

11. Two cylinders A and B fitted with pistons contain equal amounts of an ideal diatomic gas at 300 K. The piston A is free to move while that of B is held fixed. The same amount of heat is given to the gas in each cylinder. If the rise in temperature of the gas in A is 30 K, then the rise in temperature of the gas in B is

(A) 30 K (B) 18 K
(C) 50 K (D) 42 K

12. In a room where temperature is 30°C, a body cools from 61°C to 59°C in 4 minutes. The time taken by the body to cool from 51°C to 49°C will be

(A) 4 min (B) 6 min
(C) 5 min (D) 8 min

***Space for Rough work***

13. The motion of a particle varies with time according to the relation
$$y = a(\sin\omega t + \cos\omega t)$$
(A) the motion is oscillatory but not SHM
(B) the motion is SHM with amplitude a
(C) the motion is SHM with amplitude $a\sqrt{2}$
(D) the motion is SHM with amplitude 2a

14. If x, v and a denote the displacement, the velocity and the acceleration of a particle executing SHM of time period T, then which of the following does not change with time?
(A) $3a^2T^2 + 4\pi^2v^2$ (B) $\frac{aT}{x}$
(C) $aT + 2\pi v$ (D) $\frac{aT}{v}$

15. A sound wave of wavelength $\lambda$ travels towards horizontally with a velocity v. It strikes and reflects from a vertical plane surface, traveling at a speed $v_0$ towards the left. The number of positive crests striking in a time interval of 3 sec. on the wall is
(A) $\frac{3(v+v_0)}{\lambda}$ (B) $\frac{3(v-v_0)}{\lambda}$
(C) $\frac{(v+v_0)}{3\lambda}$ (D) $\frac{(v-v_0)}{3\lambda}$

16. A section of an infinite rod of charge having linear charge density $\lambda$ which is constant for all points on the line. Electric field E at a distance r from the line is
(A) $\frac{\lambda}{\pi\varepsilon_0 r}$ (B) $\frac{\lambda}{2\pi\varepsilon_0 r}$
(C) $\frac{\lambda}{4\pi\varepsilon_0 r}$ (D) $\frac{2\lambda}{\pi\varepsilon_0 r}$

17. Potential in the x-y plane is given as $V = 5(x^2 + xy)$ volts. The electric field at the point (1,-2) will be
(A) $3\hat{j}$ V/m (B) $-5\hat{j}$ V/m
(C) $5\hat{j}$ V/m (D) $-3\hat{j}$ V/m

***Space for Rough work***

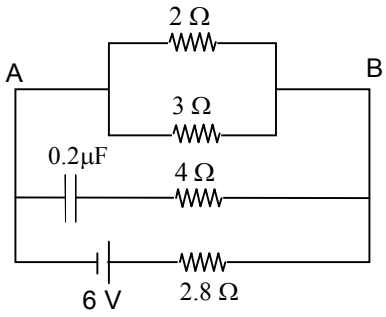
18. What is the mechanical work done in pulling the slab out of the capacitor after disconnecting it from the battery?

(A) $\frac{1}{2}E^2C(\varepsilon_r - 1)$ (B) $E^2C$

(C) $E^2C(\varepsilon_r - 1)$ (D) none of these

19. Calculate the steady-state current in the $2\Omega$ resistor shown. The internal resistance of the battery is negligible and the capacitance of the capacitor is $0.2\ \mu F$.

(A) 0.6 A
(B) 0.7 A
(C) 0.8 A
(D) 0.9 A




20. In the secondary circuit of a potentiometer a cell of internal resistance $1.5\ \Omega$ gives a balancing length of 52 cm. To get a balancing length of 40 cm, how much resistance is to be connected across the cell?

(A) $5\ \Omega$. (B) $6\ \Omega$.

(C) $7\ \Omega$. (D) $4\ \Omega$.

21. The mass of the three wire of copper are in the ratio 1 : 3 : 5, and their lengths are in ratio 5 : 3 : 1. The ratio of their electrical resistance is

(A) 1 : 3 : 5 (B) 5 : 3 : 1

(C) 1 : 15 : 125 (D) 125 : 15 : 1

22. A point P lies on the axis of a flat coil carrying a current. The magnetic moment of the coil is $\mu$. The distance of P from the coil is d, which is large compared to the radius of the coil. The magnetic field at P has magnitude

(A) $\frac{\mu_0}{2\pi}\left(\frac{\mu}{d^3}\right)$ (B) $\frac{\mu_0}{4\pi}\left(\frac{\mu}{d^3}\right)$

(C) $\frac{\mu_0}{6\pi}\left(\frac{\mu}{d^2}\right)$ (D) $\frac{\mu_0}{8\pi}\left(\frac{\mu}{d^2}\right)$

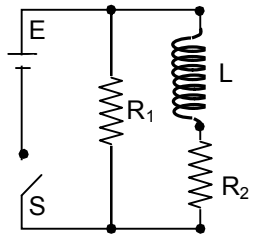
***Space for Rough work***

23. A winding wire which is used to frame a solenoid can bear a maximum 10 A current. If length of solenoid is 80 cm, then length of its winding wire is (B = 0.2 T, r = $3 \times 10^{-2}$m)

(A) $1.2 \times 10^2$ m (B) $4.8 \times 10^2$ m

(C) $2.4 \times 10^3$ m (D) $6 \times 10^3$ m

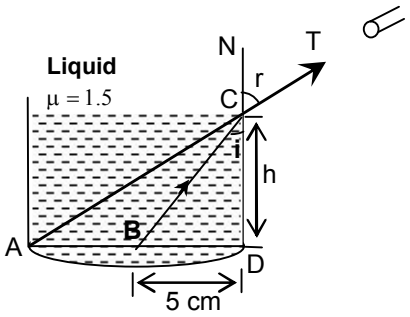
24. An inductor of inductance L = 400 mH and resistors of resistance $R_1 = 2\,\Omega$ and $R_2 = 2\,\Omega$ are connected to a battery of emf 12 V as shown in the figure. The internal resistance of the battery is negligible. The switch S is closed at t = 0. The potential drop across L as a function of time is




(A) $6e^{-5t}$ V (B) $\frac{12}{t}e^{-3t}$ V

(C) $6\left(1-e^{-t/0.2}\right)$ V (D) $12e^{-5t}$ V

25. A person looking through a telescope T just sees the point A on the rim at the bottom of a cylindrical vessel when the vessel is empty. When the vessel is completely filled with a liquid $(\mu = 1.5)$, he observes a mark at the centre, B, of the bottom without moving telescope or the vessel. What is the height of the vessel if the diameter of its cross-section is 10 cm?




(A) 5.5 cm
(B) 6.6 cm
(C) 7.5 cm
(D) 8.45 cm

26. The focal length of a plano convex lens when curved surface is silvered and the object is in front of plane surface is

(A) $\frac{R}{2\mu}$ (B) $\frac{2R}{\mu}$

(C) $-\frac{R}{2\mu}$ (D) $-\frac{2R}{\mu}$

***Space for Rough work***

27. At what distance from the objective should an object be placed to focus it properly so that the final image is formed at the least distance of clear vision?
(A) 1.1 cm (B) 2.2 cm
(C) 3.3 cm (D) 4.4 cm

28. The wavelength of 10 keV electron beam is 0.1227 $A^o$. when these waves are diffracted from a metal foil having d = 0.55 $A^o$ the first maxima would occur at an angle $\theta$ where $\sin\theta$ is
(A) 0.1116 (B) 0.2218
(C) 0.4464 (D) 0.8928

29. The wavelength of the photoelectric threshold for silver is $\lambda_0$. The energy of the electron ejected from the surface of silver by an incident light of wavelength $\lambda(\lambda < \lambda_0)$ will be
(A) $hc(\lambda_0 - \lambda)$ (B) $\dfrac{hc}{\lambda_0 - \lambda}$
(C) $\dfrac{h}{c}\left(\dfrac{1}{\lambda} - \dfrac{1}{\lambda_0}\right)$ (D) $hc\left(\dfrac{\lambda_0 - \lambda}{\lambda_0\lambda}\right)$

30. If resistivity of pure silicon is $3000\Omega m$ and the electron and hole mobilities are $0.12\ m^2V^{-1}s^{-1}$ and $0.045\ m^2V^{-1}s^{-1}$ respectively. The resistivity of a specimen of the material when $10^{19}$ atoms of phosphorous are added per $m^3$ is
$\left(\text{Given } e = 1.6\times10^{-19}c,\ \rho=3000\Omega m,\ \mu_e = 0.12\ m^2V^{-1}s^{-1},\ \mu_h = 0.045\ m^2V^{-1}s^{-1}\right)$
(A) $2.21\ \Omega m$ (B) $3.21\ \Omega m$
(C) $4.21\ \Omega m$ (D) $5.21\ \Omega m$

***Space for Rough work***

# Chemistry

## PART – II

## SECTION – A

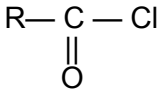
**Single Correct Choice Type**

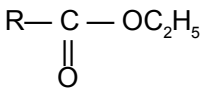
This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

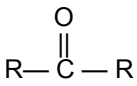
1. In the given reaction,

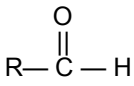
$$\underset{\text{(excess)}}{(A)} + RMgX \xrightarrow{H^+/H_2O} 3^\circ \text{ Alcohol}$$

(A) cannot be

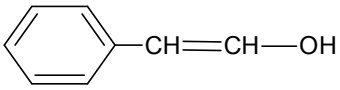
(A) 

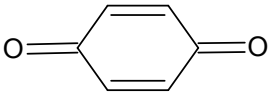
(B) 

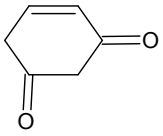
(C) 

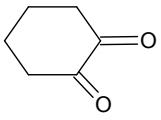
(D) 

2. Tautomerism is not exhibited by

(A) 

(B) 

(C) 

(D) 

3. Match the following:

| Column-I | | Column-II | |
|---|---|---|---|
| (A) | Boric acid | (P) | $sp^2$ hybridised boron |
| (B) | Diborane | (Q) | Ring having $6\pi$ $e^-$ delocalised |
| (C) | Borazine | (R) | Reacts with water |
| (D) | Boron Nitride | (S) | Bond having partial double bond character |
| | | (T) | $sp^3$ hybridised boron |

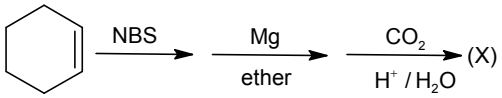
| | | | | |
|---|---|---|---|---|
| (A) | A→PRS | B→RT | C→PQRS | D→PQRS |
| (B) | A→PR | B→PR | C→PQR | D→PQR |
| (C) | A→RS | B→RT | C→PQRS | D→PQRS |
| (D) | A→RST | B→PRS | C→PQR | D→PRS |

***Space for Rough work***

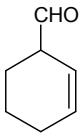
4. $I_2(s) \mid I^-$ (0.1M) half cell is connected to a $H^+$ (aq) | $H_2$ (1 bar) | Pt half cell and e.m.f. is found to be 0.7714V. If $E^o_{I_2/I^-} = 0.535V$, Then the pH of $H^+/H_2$ half cell is :

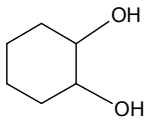
(A) 1 (B) 3
(C) 5 (D) 7

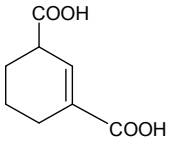
5. In the given reaction

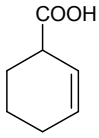



(X) will be :

(A) 


(B) 


(C) 


(D) 


6. Equivalent mass of $As_2S_3$ and $KMnO_4$ during the formation of $H_3AsO_4$, $H_2SO_4$ and $Mn^{2+}$ ions are respectively :

(A) M/2, M/5 (B) M/6, M/5
(C) M/28, M/5 (D) M /14, M/3

7. What will be the maximum value of $\Sigma$ (n + $\ell$ + m) for all unpaired electrons of chlorine in its second excited state ?

(A) 25 (B) 20
(C) 28 (D) 27

8. A gaseous compound A reacts by three independent first order processes with rate constant
$2\times10^{-3}$, $3\times10^{-3}$ and $1.93\times10^{-3}$ $sec^{-1}$ for products B, C and D respectively. If initial pressure of pure A was 8 atm taken in the closed container. Then the partial pressure of B (in atm) after 100 sec from starting:

(A) 0.288 (B) 0.577
(C) 1.154 (D) 2.15

***Space for Rough work***

9. Which of the following statements for crystals having frenkel defect is not correct?
(A) Frenkel defects are observed where the difference in sizes of cations and anions is large
(B) The density of crystals having frenkel defect is lesser than that of a pure perfect crystal
(C) In an ionic crystal having frenkel defect may also contain schottky defect
(D) Usually alkali halides do not have frenkel defect

10. Which of the following is true in respect of chemical adsorption?
(A) $\Delta H < 0, \Delta S > 0, \Delta G > 0$ (B) $\Delta H > 0, \Delta S > 0, \Delta G < 0$
(C) $\Delta H < 0, \Delta S < 0, \Delta G < 0$ (D) $\Delta H > 0, \Delta S < 0, \Delta G > 0$

11. 0.1 M NaCl and 0.05 M $BaCl_2$ solutions are separated by a semi-permeable membrane in a container. For this system, choose the correct answer :
(A) There is no movement of any solution across the membrane
(B) Water flows from $BaCl_2$ solution toward NaCl solution
(C) Water flows from NaCl solution towards $BaCl_2$ solution
(D) Osmotic pressure of 0.1M NaCl is lower than the osmotic pressure of $BaCl_2$

12. Glucose reacts with acetic anhydride to form :
(A) hexaacetate (B) tetraacetate
(C) monoacetate (D) pentaacetate

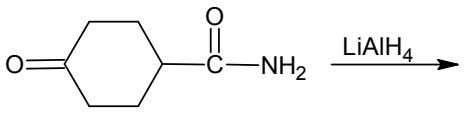
13. Among the following the coloured compound is :
(A) CuCl (B) $K_3[Cu(CN)_4]$
(C) $CuF_2$ (D) $[Cu(CH_3CN)_4]\ BF_4$

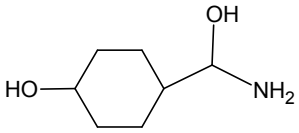
14. Identify the incorrect statement :
(A) Liquid oxygen is an important constituent of rocket fuel
(B) When $SO_2$ gas is passed through cupric chloride solution, then the solution becomes colourless and a white precipitate is formed
(C) Radon is formed from the decay of Radium
(D) Hydrofluoric acid is preserved in glass bottles

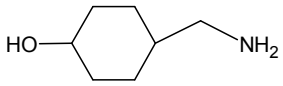
15. Which of the following ores is best concentrated by froth floatation process:
(A) Malachite (B) Cassiterite
(C) Galena (D) Magnetite

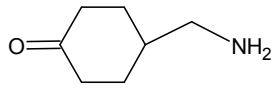
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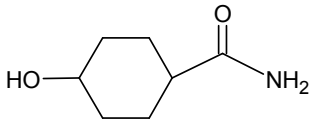
16. The IUPAC name of $[Ni(NH_3)_4][NiCl_4]$ is:
(A) Tetrachloronickel (II) tetra ammine nickel (II)
(B) Tetraammine nickel (II) tetra chloronickelate (II)
(C) Tetraamminenickel (II) tetrachloronickel (II)
(D) Tetrachloronickel (II) tetra amminenickelate (O)

17. 

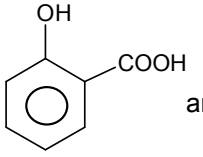

(A) 

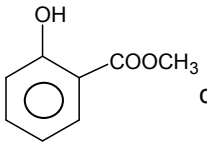

(B) 


(C) 


(D) 


18. Which of the following is not correctly matched ?
(A) Phenol + $CHCl_3$ + NaOH $\xrightarrow{\text{heat}}$ Salicylaldehyde
(B) Phenol + Phthalic anhydride $\xrightarrow[H_2SO_4]{\text{heat}}$ Phenetole
(C) $C_6H_5-\overset{+}{N}\equiv NCl^- \xrightarrow[H_2O]{H_3PO_2} C_6H_6$
(D) $C_6H_6 + CH_3COCl \xrightarrow[AlCl_3]{\text{anhyd}} C_6H_5COCH_3 + HCl$

19. 

and 

can be differentiated by :
(A) NaOH
(B) Na metal
(C) $NaHCO_3$
(D) $FeCl_3$

20. The monomers of Buna-S polymers are :
(A) Vinyl chloride and Vinylidene
(B) Styrene and butadiene
(C) Acrylonitrile and butadiene
(D) Isobutylene and isoprene

***Space for Rough work***

21. In the reaction, $4A + 2B + 3C \rightarrow A_4B_2C_3$, what will be the number of moles of product formed, starting from 1 mol of A, 0.6 mol of B and 0.72 mol of C?
(A) 0.25 (B) 0.3
(C) 0.24 (D) 2.32

22. The incorrect statement among the following is
(A) In $T\ell I_3$ the oxidation state of $T\ell$ is +3.
(B) $B_2O_3$ reacts with $P_2O_5$ to form $BPO_4$.
(C) $B(OH)_3$ solution is acidic in nature.
(D) Borax solution can be used as a buffer.

23. Sodium oxide $Na_2O$ is a crystalline solid. It does not have.
(A) Antiflourite structure
(B) $Na^+$ ions are present at body diagonals
(C) $Na^+$ ions are present at tetrahedral voids
(D) $O^{2-}$ ions are present at octahedral voids

24. The oxidation number of carbon in H — C ≡ N and H – N $\overrightarrow{=}$ C are respectively :
(A) +2, +3 (B) +2, +4
(C) +2, +2 (D) +3, +3

25. 0.56 litre each of three samples of $H_2O_2$ labelled as 10 vol, 15 vol and 20 vol are mixed. The resultant normality of $H_2O_2$ solution is :
(A) 1.68 (B) 2.68
(C) 4.5 (D) 8.0

26. Calculate the vapour pressure of water at 300K if its heat of vaporisation is 540 cal/gm
(A) 3.162 mm (B) 0.3192 mm
(C) 31.92 mm (D) 319.2 mm

27. Select the incorrect statement :
(A) $O_3$ and $O^{2-}$ both are diamagnetic
(B) Out of $O_2$, $O_2^+$, $O_3$ , least O – O bond length is in $O_2^+$
(C) Out of $O_2$, $O_2^+$, $O_2^-$, only $O_2$ is paramagnetic
(D) Out of $O_2$, $O_2^+$, $O_2^-$, maximum spin magnetic moment have $O_2$

***Space for Rough work***

28. Which systematic diagram represents the correct sequence of reaction?

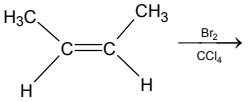
(A) $CaCO_3 \xrightarrow{\Delta} CO_2 + CaO$; $CaO \xrightarrow{H_2O} Ca(OH)_2$; $Ca(OH)_2 \xrightarrow{CO_2} CaCO_3$

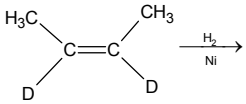
(B) $CaCO_3 \xrightarrow{\Delta} CO_2 + CaO$; $CaO \xrightarrow{H_2O} Ca(OH)_2$; $Ca(OH)_2 \xrightarrow{\text{excess of } CO_2} CaCO_3$

(C) $CaCO_3 \xrightarrow{\Delta} CaO + CO_2$; $CaO \xrightarrow{H_2O\ (cold)} H_2CO_3$; $H_2CO_3 \xrightarrow{CaCl_2 + HCl(aq)} CaCO_3$

(D) $CaCO_3 \xrightarrow{\Delta} CO_2 + CaO$; $CaO \xrightarrow{H_2O} Ca(OH)_2$; $Ca(OH)_2 \xrightarrow{SO_2} CaCO_3$

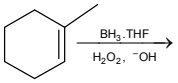
29. How many of the following addition reactions are syn addition reaction:

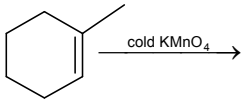
(a) 

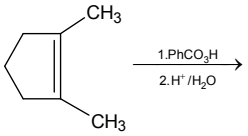
(b) 

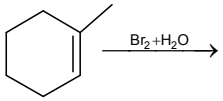
(c) $CH_3 - C \equiv C - CH_3 \xrightarrow[BaSO_4]{H_2, Pd}$

(d) $CH_3 - C \equiv C - CH_3 \xrightarrow{Na,\ liq.NH_3}$

(e) 

(f) 

(g) 

(h) 

(A) 3
(B) 6
(C) 5
(D) 4

30.

| | Column-I | Column-II |
|---|---|---|
| (A) |  | (P) Hyperconjugation |
| (B) |  | (Q) All carbon atom are $sp^2$ hybridized |
| (C) |  | (R) Aromatic |
| (D) | $(CH_3)_3 - C^{\oplus}$ | (S) Diamagnetic |

(A) A→PQR B→PQR C→PQR D→PQR
(B) A→QRS B→QRS C→RS D→PQR
(C) A→QRS B→QRS C→QRS D→P,S
(D) A→QRS B→QS C→QR D→P,S

***Space for Rough work***

# *Mathematics* PART – III

## SECTION – A
### Single Correct Choice Type

This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. A and B throw a pair of dice alternately till one of them wins. If A wins by throwing 7 while B wins by throwing 5 as a sum respectively, then the probability that one of the dice shows 3 on the last throw if A starts the game is

   (A) $\frac{11}{36}$ (B) $\frac{11}{28}$

   (C) $\frac{7}{22}$ (D) none of these

2. The sum of rational term(s) in $\left(\sqrt{3}+\sqrt[3]{2}+\sqrt[4]{5}\right)^8$ is equal to

   (A) 3150 (B) 336

   (C) 3486 (D) 3592

3. $e^{|\sin x|}+e^{-|\sin x|}+4a=0$ will have exactly four different solutions in $[0, 2\pi]$ if

   (A) $a \in \left[-\frac{e}{4}, -\frac{1}{4}\right]$ (B) $a \in R$

   (C) $a \in \left[\frac{-1-e^2}{4e}, \infty\right)$ (D) none of these

4. If $2^{\frac{2\pi}{\sin^{-1}x}}-2(a+2)2^{\frac{\pi}{\sin^{-1}x}}+8a<0$ for at least one real x, then

   (A) $a \in \left[-\infty, \frac{1}{8}\right) \cup (2, \infty)$ (B) $a \in R - \{2\}$

   (C) $a < 2$ (D) $\frac{1}{8} \le a < 2$

***Space for rough work***

5. Let f(x) = $\sin x\left(\cos x+\sqrt{\cos^2 x+\cos^2\theta}\right)$, where '$\theta$' is a given constant, then maximum value of f(x) is

(A) $\sqrt{1+\cos^2\theta}$ (B) $|\cos\theta|$

(C) $|\sin\theta|$ (D) $\sqrt{1+\sin^2\theta}$

6. The number of integral values of K for which the equation 7 cos x + 5 sin x = 2k + 1 has a solution

(A) 5 (B) 6

(C) 7 (D) 8

7. If $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \pi$, then $x^4 + y^4 + z^4 + 4x^2y^2z^2 = k(x^2y^2 + y^2z^2 + z^2x^2)$, where k is equal to

(A) 1 (B) 2

(C) 4 (D) none of these

8. If the variable takes values 0, 1, 2, 3, ….., n with frequencies proportional to ${}^nC_0$, ${}^nC_1$, ${}^nC_2$, ….., ${}^nC_n$ respectively, the variance is

(A) $\frac{n}{3}$ (B) $\frac{2n}{5}$

(C) $\frac{n}{4}$ (D) none of these

9. Let a relation R in the set N of natural numbers be defined as (x, y) $\Leftrightarrow x^2 - 8xy + 7y^2 = 0$ $\forall$ x,
y $\in$ N. The relation R is

(A) reflexive (B) symmetric

(C) transitive (D) none of these

10. Suppose $f(x) = x^3 + ax^2 + 2bx + c$, where a, b, c are chosen respectively by throwing a dice three times. Then the probability that f(x) is a decreasing function for some x is

(A) $\frac{1}{3}$ (B) $\frac{13}{36}$

(C) $\frac{2}{9}$ (D) none of these

***Space for rough work***

11. A line intersects the ellipse $\frac{x^2}{4a^2}+\frac{y^2}{a^2}=1$ at A and B and the parabola $y^2 = 4a(x + 2a)$ at C and D. The line segment AB subtends a right angle at the centre of the ellipse. Then, the locus of the point of intersection of tangents to the parabola of C and D is

(A) $y^2 - a^2 = \frac{5}{4}(x-4a)^2$ (B) $y^2 - 2a^2 = 10(x - 4a)^2$

(C) $y^2 + a^2 = \frac{5}{2}(x-4a)^2$ (D) $y^2 + 4a^2 = 5(x + 4a)^2$

12. If $f(x)=\frac{9^x}{9^x+9}$, then $f\left(\frac{1}{2015}\right)+f\left(\frac{2}{2015}\right)+f\left(\frac{3}{2015}\right)+.....+f\left(\frac{4029}{2015}\right)$ is equal to

(A) 1007 (B) $\frac{4029}{2}$

(C) 2014 (D) 2015

13. If $p_i=\frac{|x-a_i|}{x-a_i}$, (where i = 1, 2, 3, ….., p) and $a_1 > a_2 > a_3$ ….. $a_p$, then $\lim\limits_{x\to a_q}\left(p_1p_2 ..... p_p\right)$, $1 \le q \le p$ is equal to

(A) $(-1)^{q+1}$ (B) $(-1)^{q-1}$

(C) $(-1)^q$ (D) does not exist

14. $\int\sqrt{x-3}\left(\sin^{-1}(\ln x)+\cos^{-1}(\ln x)\right)dx$ is equal to

(A) $\frac{\pi}{3}(x-3)^{3/2}+C$ (B) 0

(C) DNE (D) none of these

15. If $x\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$, then the value of $\tan^{-1}\left(\frac{\tan x}{4}\right)+\tan^{-1}\left(\frac{3\sin 2x}{5+3\cos 2x}\right)$ is

(A) 2x (B) 3x

(C) $\frac{x}{2}$ (D) x

***Space for rough work***

16. P is any point on the circum-circle of $\Delta ABC$ other than the vertices. H is the orthocentre of $\Delta ABC$, M is the mid-point of PH and D is the mid-point of BC. Then,
(A) DM is parallel to AP (B) DM is perpendicular to AP
(C) AP is opposite side of DM (D) none of these

17. Let $f(x) = (x-3)^3 x^n$, $n \in N$. Then, f(x) has a
(A) minimum at x = 0, if n is even (B) maximum at x = 0, if n is even
(C) maximum at $x = 3\ \forall\ n \in N$ (D) none of these

18. If a function y = f(x) is defined as $y = \frac{1}{t^2 - t - 6}$ and $t = \frac{1}{x-2}$, $t \in R$. Then, f(x) is discontinuous at
(A) $2, \frac{2}{3}, \frac{7}{3}$ (B) $2, \frac{3}{2}, \frac{7}{3}$
(C) $2, \frac{3}{2}, \frac{5}{3}$ (D) none of these

19. $\int \left(1 + x - x^{-1}\right) e^{x+x^{-1}} dx$ is equal to
(A) $x \cdot e^{x+x^{-1}}$ (B) $-x \cdot e^{x+x^{-1}} + c$
(C) $(x+1)e^{x+x^{-1}} + c$ (D) $(x-1)e^{x+x^{-1}} + c$

20. The co-ordinates of four vertices of a quadrilateral are (–2, 4), (–1, 2), (1, 2) and (2, 4) taken in order. The equation of the line passing through the vertex (–1, 2) and dividing the quadrilateral in two equal areas is
(A) x + 1 = 0 (B) x + y = 1
(C) x – y + 3 = 0 (D) none of these

21. $\sum_{r=0}^{n} (-1)^r\ {}^nC_r \frac{1 + r\log_e 10}{\left(1 + \log_e 10^n\right)^r}$ equals
(A) 1 (B) –1
(C) n (D) none of these

22. If a, b, c are distinct positive numbers such that b + c – a, c + a – b and a + b – c are positive, then the expression (b + c – a)(c + a – b)(a + b – c) – abc is
(A) positive (B) negative
(C) non-positive (D) non-negative

***Space for rough work***

23. PSQ is a focal chord of a parabola whose focus is S and vertex is A. PA, QA are produced to meet the directrix in R and T. Then angel $\angle$RST is equal to
(A) 90º (B) 60º
(C) 45º (D) 30º

24. If $z_1$ and $z_2$ are lying on $|z - 3| \le 4$ and $|z - 1| + |z + 1| = 3$ respectively. Then, $A = |z_1 - z_2|$ satisfies
(A) $0 \le A < \frac{15}{2}$ (B) $0 < A \le \frac{15}{2}$
(C) $0 \le A \le \frac{17}{2}$ (D) $0 < A < \frac{17}{2}$

25. If four points be taken on a rectangular hyperbola such that the chord joining any two is perpendicular to the chord joining the other two, and if $\alpha$, $\beta$, $\gamma$, $\delta$ be the inclinations to either asymptote of the straight line joining these points to the centre, then $\tan\alpha \tan\beta \tan\gamma \tan\delta$ is equal to
(A) 1 (B) 2
(C) –2 (D) –1

26. The sum of squares of the perpendiculars drawn from the points (0, 1) and (0, –1) to any tangent to a curve is 2. Then, the equation of the curve is
(A) $2y = c(x + 2)$ (B) $y = c(x \pm 1)$
(C) $y = c(x + 2)$ (D) $y = c(x \pm 2)$

27. The area of the region bounded by the curves $y = x^2$, $y = |2 - x^2|$ and $y = 2$ which lies to the right of the line $x = 1$, is
(A) $\left(\frac{12 - 20\sqrt{3}}{2}\right)$ sq. units (B) $\left(\frac{20 - \sqrt{2}}{3}\right)$ sq. units
(C) $\left(\frac{20 - 12\sqrt{2}}{3}\right)$ sq. units (D) $\left(\frac{12 - 20\sqrt{2}}{3}\right)$ sq. units

***Space for rough work***

28. If A, B, C are angles of a triangle, then $\begin{vmatrix} e^{2iA} & e^{-iC} & e^{-iB} \\ e^{-iC} & e^{2iB} & e^{-iA} \\ e^{-iB} & e^{-iA} & e^{2iC} \end{vmatrix}$ is equal to

(A) 4 (B) 0
(C) –4 (D) –1

29. Two persons who are 500 m apart, observe the direction and the angle of elevation of a balloon at the same instant. One finds the elevation to be $\frac{\pi}{3}$ and direction being South-West, while the other finds the elevation to be $\frac{\pi}{4}$ and direction being west. Height of the balloon is

(A) $500\sqrt{\frac{3}{4-\sqrt{6}}}$ m (B) $250\sqrt{\frac{3}{4+\sqrt{6}}}$ m
(C) $250\sqrt{\frac{3}{4-\sqrt{6}}}$ m (D) $500\sqrt{\frac{3}{4+\sqrt{6}}}$ m

30. Let p and q be two statements, then $\sim(\sim p \wedge q) \wedge (p \vee q)$ is logically equivalent to
(A) q (B) $p \wedge q$
(C) p (D) $p \vee \sim q$

***Space for rough work***